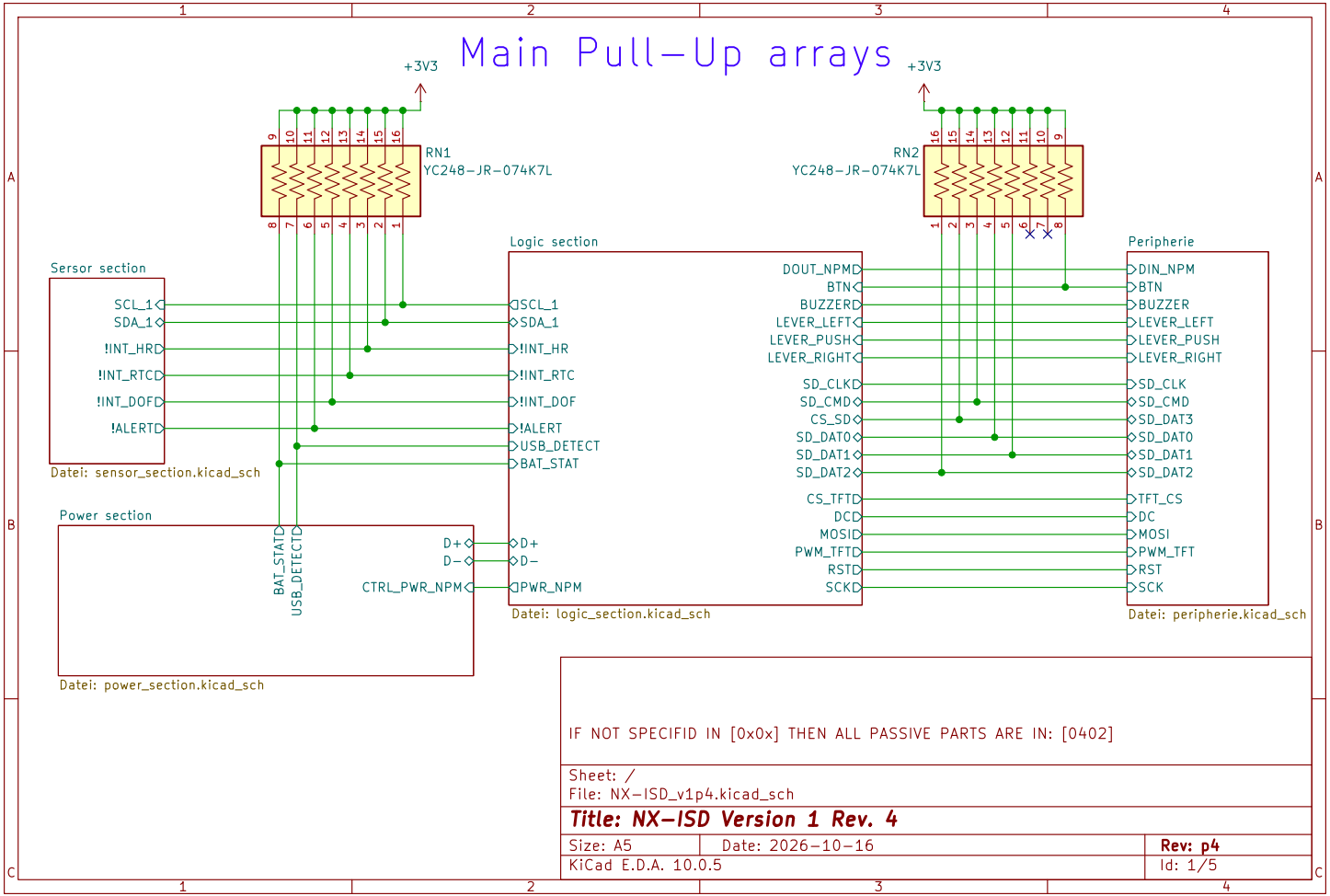
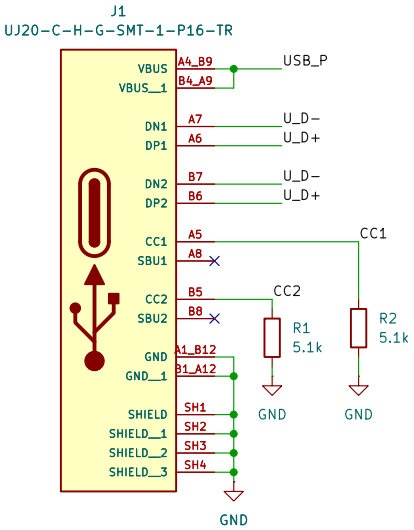
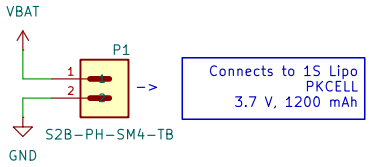

Main Pull-Up arrays
+3V3
RN1
YC248-JR-074K7L
+3V3
RN2
YC248-JR-074K7L
Sersor section
SCL_1
SDA_1
!INT_HRD
!INT_RTCD
!INT_DOFD
!ALERTD
Datei: sensor_section.kicad_sch
Logic section
SCL_1
SDA_1
!INT_HR
!INT_RTC
!INT_DOF
!ALERT
USB_DETECT
BAT_STAT
DOUT_NPM
BTN
BUZZERD
LEVER_LEFT
LEVER_PUSH
LEVER_RIGHT
SD_CLK
SD_CMD
CS_SD
SD_DAT0
SD_DAT1
SD_DAT2
CS_TFT
DC
MOSI
PWM_TFT
RST
SCK
D+
D-
PWR_NPM
Datei: logic_section.kicad_sch
Peripherie
DIN_NPM
BTN
BUZZER
LEVER_LEFT
LEVER_PUSH
LEVER_RIGHT
SD_CLK
SD_CMD
SD_DAT3
SD_DAT0
SD_DAT1
SD_DAT2
TFT_CS
DC
MOSI
PWM_TFT
RST
SCK
Datei: peripherie.kicad_sch
Power section
BAT_STAT
USB_DETECT
D+
D-
CTRL_PWR_NPM
Datei: power_section.kicad_sch
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]
Sheet: /
File: NX-ISD_v1p4.kicad_sch
Title: NX-ISD Version 1 Rev. 4
Size: A5
Date: 2026-10-16
Rev: p4
KiCad E.D.A. 10.0.5
Id: 1/5

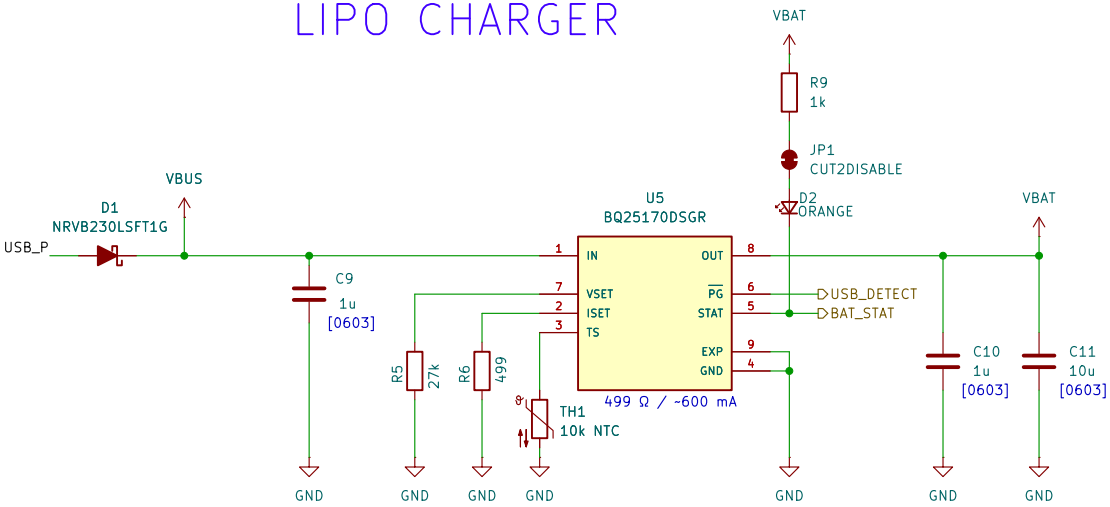
USB-C & LiPo CONNECTOR





Connects to 1S Lipo
PKCELL
3.7 V, 1200 mAh

LIPO CHARGER

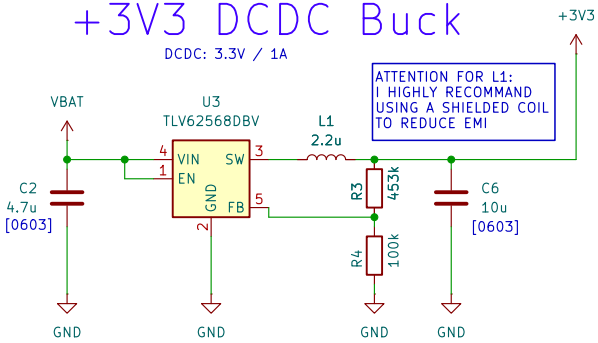


Formula R_ISET: 300/ I_OUT(in Amperes)

Charge Current	Calculated R_ISET
50 mA	6000 Ω
100 mA	3000 Ω
150 mA	2000 Ω
200 mA	1500 Ω
300 mA	1000 Ω
400 mA	750 Ω
500 mA	600 Ω
600 mA	500 Ω
800 mA	375 Ω

+3V3 DCDC Buck

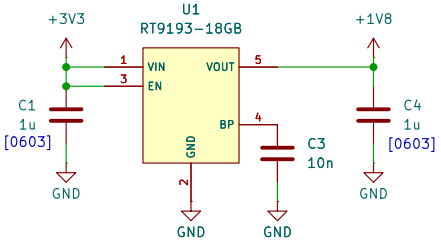
DCDC: 3.3V / 1A



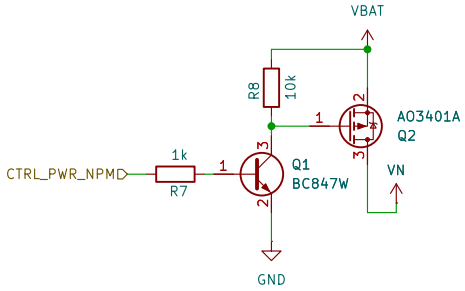
ATTENTION FOR L1:
I HIGHLY RECOMMAND
USING A SHIELDED COIL
TO REDUCE EMI

+1V8 LDO

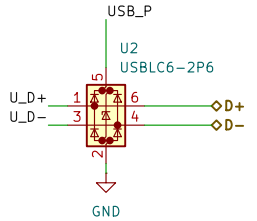
(USED FOR HRM)



NPM POWER CONTROL



ESD SAFETY DIODES



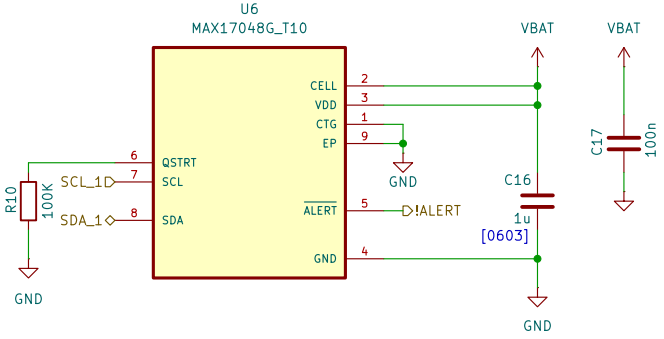
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Power section/
File: power_section.kicad_sch

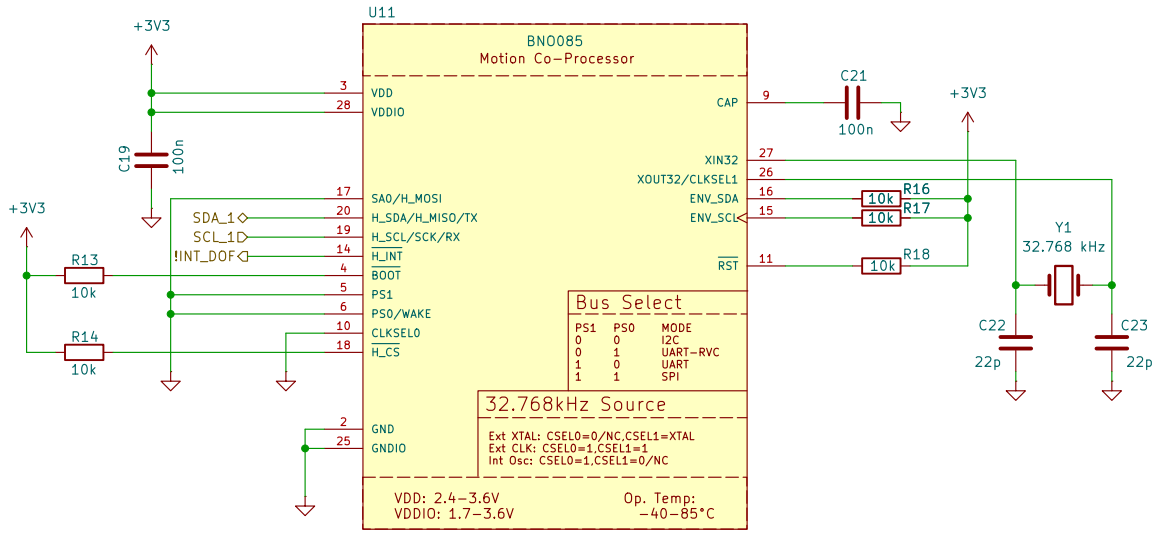
Title: NX-ISD Version 1 Rev. 4

Size: A4	Date: 2026-10-16	**Rev: p4**
KiCad E.D.A. 10.0.5		Id: 2/5

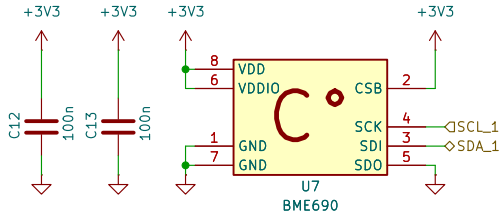
Battery Fuel Gauge



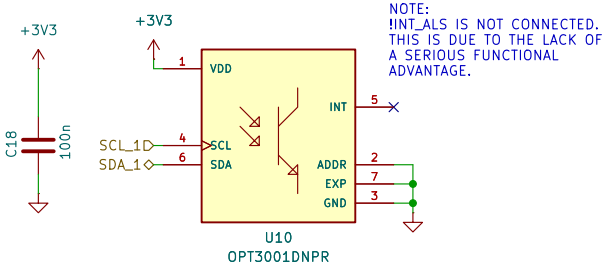
9-DOF



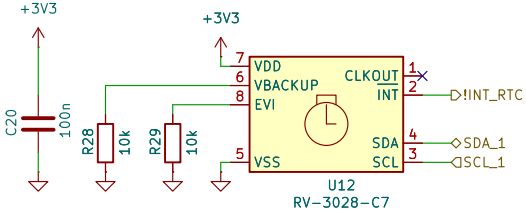
Temp/Humidity/Pressure



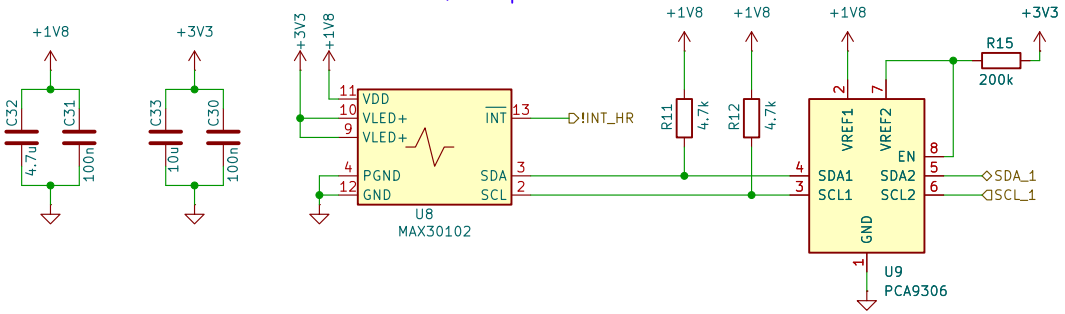
Ambient Light Sensor



RTC



BPM/SpO2



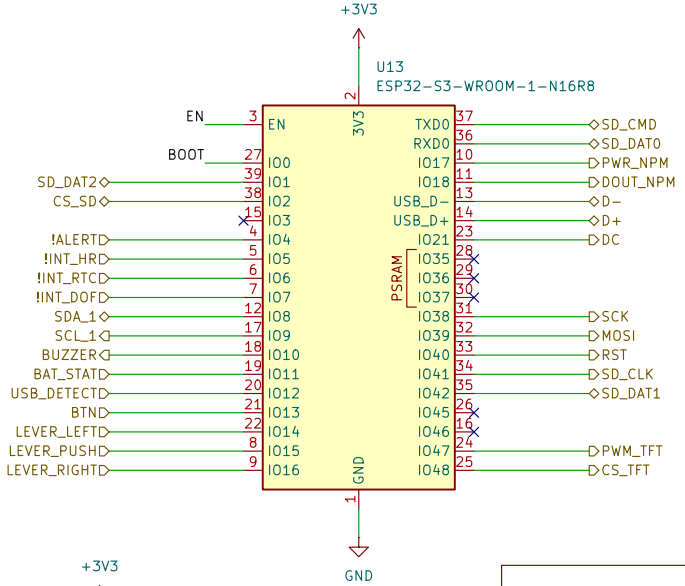
IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Sersor section/
File: sensor_section.kicad_sch

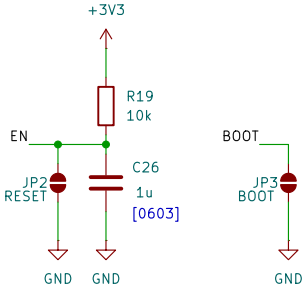
Title: NX-ISD Version 1 Rev. 4

Size: A4	Date: 2026-10-16	**Rev: p4**
KiCad E.D.A. 10.0.5		Id: 3/5

ESP32-S3 N16R8

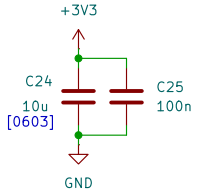


EN & BOOT



ESP32-S3 chip strapping pins:

GPIO 0.
GPIO 3.
GPIO 45.
GPIO 46.



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

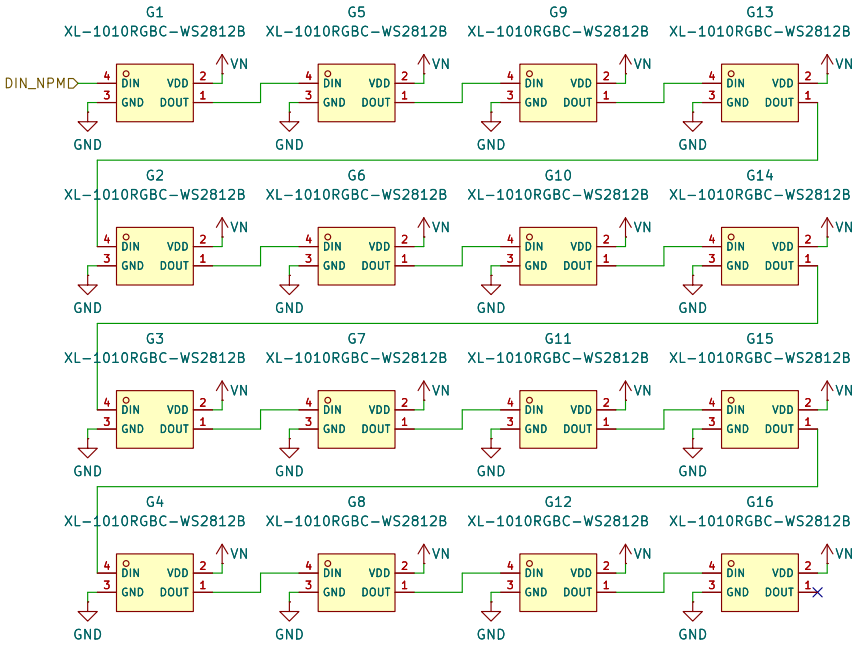
Sheet: /Logic section/
File: logic_section.kicad_sch

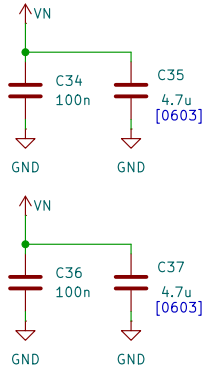
Title: NX-ISD Version 1 Rev. 4

Size: A5	Date: 2026-10-16	**Rev: p4**
KiCad E.D.A. 10.0.5		Id: 4/5

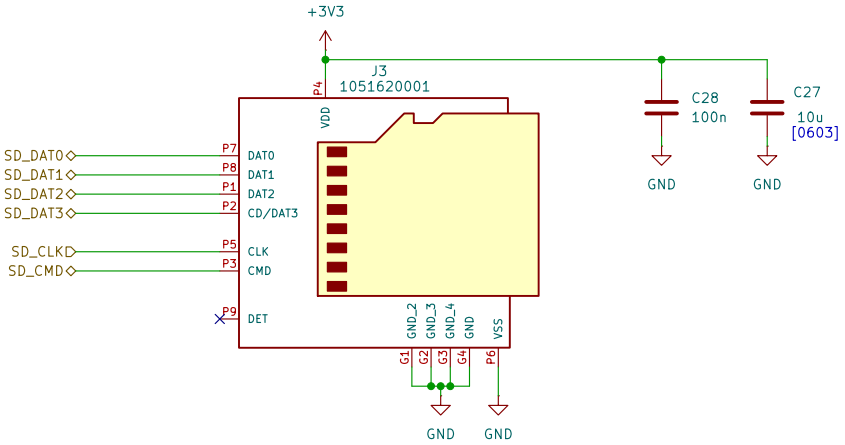
NeoPixel Matrix (NPM)

(FOR STABALIZING THE NPM RECEIVES TWO BULK CAPACITOR PAIRS ON LEFT AND RIGHT)

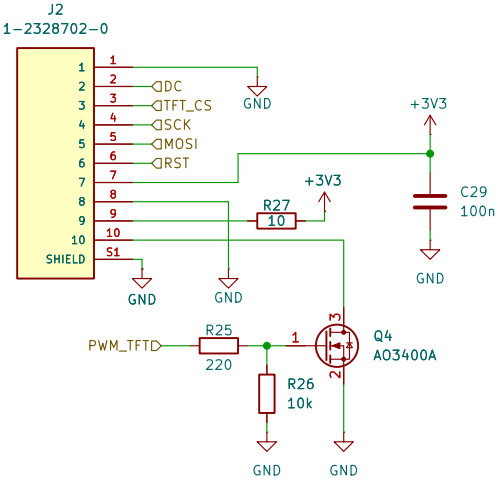




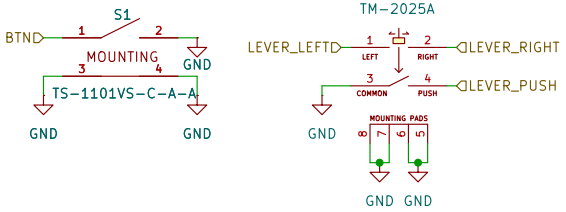
uSD-CARD SLOT



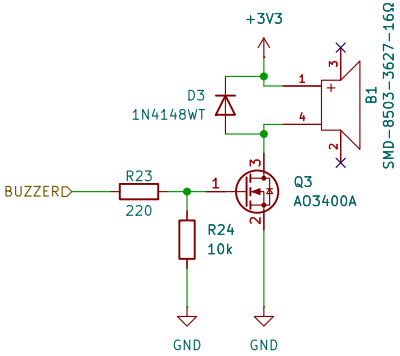
DISPLAY



BUTTONS



BUZZER



IF NOT SPECIFID IN [0x0x] THEN ALL PASSIVE PARTS ARE IN: [0402]

Sheet: /Peripherie/
File: peripherie.kicad_sch

Title: NX-ISD Version 1 Rev. 4

Size: A4	Date: 2026-10-16	**Rev: p4**
KiCad E.D.A. 10.0.5		Id: 5/5